The Relational Emergence Hypothesis (REH): Networks, Noise, and the Interior Lining of the Universe

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Abstract

The aim of this paper is to construct, explicitly and paradox-safely, a universal set from operational materials: observer algebras and identity-safe (stratified) comprehension. It separates the ontic level (what exists) from the semantic/epistemic constructions agents use to reason about it, and demonstrates that the universe serves as the Turing-complete empirical validator for all logical claims about the physical world. We present natural-deduction (ND) derivations with every axiom/rule invocation identified. For the lemmas and theorem we provide rigorous structured proofs suitable for formalization. We discuss technical properties, model constructions, known consistency results, advantages, disadvantages, and practical recommendations for REH.

1 Introduction and Motivation

Introduction and Motivation

- **REH in one sentence:** Structure = resonance from differences in a relational field with no primitive spacetime.
- **Methodological commitment:** start from only the three axioms of classical logic; derive everything else from evidence and argument to avoid importing unstable primitives. We build the network from scratch to keep the ontology tight.
- **Why now:** persistent tensions (GR/QM, singularities, measurement, observer frames) suggest spacetime is not fundamental.
- **Three closures (to be derived, not assumed):**
  - **1. Non-uniform difference (gradients) at genesis,**
  - **2. Interior lining (observer sandbox that localizes contradiction),**
  - **3. Resonance (metastable harmonics as all "things").**

2 Logical Ground and Axioms

2.0 Logical Ground and Axioms (Gate and Filter)

Purpose of Section 2. This section is the gate/filter for the rest of the manuscript. Its goal is to make explicit the single ontic primitive we accept and to enumerate exactly which additional principles are operational choices made by agents. Nothing should be smuggled in: any extra axiom or rule used later must be traceable to a line in this section.

We adopt the following minimal commitments (decisions agreed above):

- **Root primitive:** Identity $\forall x\,(x = x)$ is the sole metaphysical datum about the ontic domain.
- **Classical logic as agent-level (Option B):** Non-contradiction and excluded middle are *conditional* choices of agents for their internal logics when their interpretations warrant it.
- **Measurement-level comprehension:** set-formation is permitted only when operationally definable via measurement/partition (or when a predicate is identity-safe/stratified; cf. Appendix B for alternatives).
- **Continuum/calculus as modelling:** allowed only where measurability/operationalization is provided; avoid arbitrary global powerset assumptions.

2.1 Ontic substrate and measurement model (Physics)

Ontic domain. Let U denote the ontic domain (collection) of physically instantiated states (microstates, macrostates, or concrete tokens). Each $u \in U$ is a brute fact ($u = u$).

Observers & measurement. Each observer/worldline o has a measurement map

$$m_o : U \to X_o,$$

where X_o is an outcome space (finite-precision reals, discrete labels, etc.). The measurement induces an observable algebra

$$\mathcal{A}_o \;=\; \sigma\big(\{\, m_o^{-1}(B) \;:\; B \subseteq X_o \,\}\big),$$

i.e., the Boolean/σ-algebra generated by the preimages of measurable outcome sets. Elements of $\mathcal{A}_o$ are the operationally reportable sets for observer o.

Quanta / referents. A quantum or referent for observer o is a block $m_o^{-1}(x)$ (or a measurable union of such blocks). Agents use these blocks as the primitive atoms for inference; those blocks are what we call the instances or member tokens of constructed sets.

Philosophical note. Physics supplies the data (measurements); it does not impose definitions beyond identity. Any claimed generalization (sets, predicates) is an interpretive layer added by agents.

2.2 Mind, interpretations, and logic (Constructed semantics)

Definition 1 (Language)**.** *Let L be an agent's object language (constants/names, function symbols, and predicate symbols of various arities), with the usual logical connectives and quantifiers.*

Definition 2 (Operational algebra)**.** *For agent i (worldline i), let $\mathcal{B}_i$ denote the Boolean (resp. σ-) algebra generated by that agent's measurable sets and any publicly externalized predicates they accept. All semantic denotations for i live in $\mathcal{B}_i$; no global powerset is assumed.*

Definition 3 (Interpretation map)**.** *An* interpretation *for agent i is a homomorphism*

$$I_i : L \to \mathcal{B}_i$$

that assigns to each constant a an element $I_i(a) \in \mathcal{B}_i$ (typically a singleton or the empty set), and to each k-ary predicate symbol P a set $I_i(P) \subseteq U^k$ whose unary projections (as used in formulas-in-context) lie in $\mathcal{B}_i$. Function symbols are handled by measurable/operational realizers whose graphs lie in $\mathcal{B}_i$.

Definition 4 (Truth under interpretation)**.** *Write $I_i \models \varphi$ when the (Tarskian) satisfaction relation holds for φ with non-logical symbols interpreted by I_i and quantification ranging over U;*[1] *connective/quantifier clauses are standard.*

Definition 5 (Validity (syntactic))**.** *Given a theory T and a set of inference rules R_i chosen by agent i (their* calculus*), say*

$$\mathrm{Valid}(T, R_i, \varphi)$$

iff there exists a formal derivation of φ from T using R_i. This is purely syntactic.

Definition 6 (Soundness (operational))**.** *T with rules R_i is* sound for agent *i iff for every ψ,*

$$\mathrm{Valid}(T, R_i, \psi) \;\Rightarrow\; I_i \models \psi.$$

Soundness is the glue between the formal arena and the ontic domain: derivable conclusions must correspond to states of affairs under I_i.

Definition 7 (Externalization)**.** *An agent* externalizes *a claim φ by publishing evidence/procedures such that other agents j can test $I_j \models \varphi$ relative to their own $\mathcal{B}_j$. Externalization promotes private denotations to intersubjectively checkable ones.*

Remark (EQ-Repl and congruence). We treat Leibniz (replacement of equals) as a *rule* of the agent's calculus R_i, not as an ontic axiom. Its admissible use requires a *congruence warrant* for the predicates/functions concerned; a soundness lemma for EQ-Repl under congruence is stated later in Section 2.

Remark (Logic pluralism). REH's ontic core is logic-agnostic. We present in classical logic for clarity, but agents may choose alternative calculi (intuitionistic, paraconsistent, etc.) provided their rules are used within $\mathcal{B}_i$ and externalized claims remain operationally testable.

[1] Quantified sets that appear extensionally must be denotable in $\mathcal{B}_i$ for the agent's uses (measurement-level comprehension).

2.3 Parsing / quantization (discretizing continuous reality)

Parsing = choosing an observable partition. For observer o, parsing a continuous manifold means selecting a partition or σ-algebra $\mathcal{A}_o$ induced by the measurement map $m_o : U \to X_o$. Each partition cell (an atom of $\mathcal{A}_o$) is a discrete *referent* used in inference: fibers $m_o^{-1}(x)$ and measurable unions thereof.

Operational constraint (measurement-level comprehension). We permit formation of sets (comprehension) only when the defining predicate is:

i) measurable in some observable algebra $\mathcal{A}_o$ for some observer o (i.e., operationalizable by measurement), or

ii) identity-safe/*stratified* (optional; see Appendix B), or

iii) explicitly definable via an agent's interpretation I_i, with a commitment to externalizable membership tests.

This is the firewall against naive unrestricted comprehension: only operationally definable (or identity-safe) sets are admitted as legitimate for inference that may be externalized to physics.

Continuum & calculus. Calculus and continuous functions are allowed as modelling tools on the manifold when they refer to measurable functions and their use is constrained by operational tolerances (finite precision, boundedness, integrability). We avoid assuming arbitrary pointwise predicates on the continuum unless they are measurable or operationally approximable.

2.4 Logic vs Physics: complementary inference validation

Definition 8 (Inference triple)**.** *An* inference operation *is a triple $\mathcal{I} = (T, R, \varphi)$ consisting of a theory T, a rule set (calculus) R, and a candidate conclusion φ.*

Define the following predicates (all agent-indexed when needed):

$$\begin{aligned}
\mathrm{Valid}(T, R, \varphi) &:\iff \varphi \text{ is derivable from } T \text{ using } R \text{ (purely syntactic).} \\
\mathrm{True}_i(\varphi) &:\iff I_i \models \varphi \quad \text{(truth under agent } i\text{'s interpretation } I_i : L \to \mathcal{B}_i). \\
\mathrm{PhysRealizable}(\mathcal{I}) &:\iff \exists \text{ operational test (in some } \mathcal{A}_o) \text{ that, within tolerances,} \\
&\qquad \text{adjudicates the premises in } T \text{ and the conclusion } \varphi.
\end{aligned}$$

Definition 9 (Observational indistinguishability)**.** *For agent i, define $u \sim_i v$ iff for all $B \in \mathcal{A}_i$ we have $(u \in B) \Leftrightarrow (v \in B)$ (equivalently, $m_i(u) = m_i(v)$ in deterministic measurement).*

Definition 10 (Congruence warrant)**.** *For agent i and a k-ary predicate symbol P, write* $\mathrm{Congruence_warrant}_i(P)$ *when, for all tuples $\boldsymbol{u}, \boldsymbol{v} \in U^k$ with $\boldsymbol{u} \sim_i \boldsymbol{v}$ (componentwise), we have*

$$P(\boldsymbol{u}) \;\Leftrightarrow\; P(\boldsymbol{v}).$$

Only under such warrants is Leibniz-style substitution (EQ-Repl) admissible for P in i's operational logic.

Proposition 1 (EQ-Repl as a rule; soundness under warrant)**.** *Leibniz replacement is treated as a* rule *in R_i, not an ontic axiom. Its applications are sound for agent i whenever the relevant predicates/functions satisfy* $\mathrm{Congruence_warrant}_i(\cdot)$.

2.4a Operational Closure and the OC-Universe

Purpose. To construct, explicitly and paradox-safely, a "universal set" from operational materials: observer algebras and identity-safe (stratified) comprehension—*without* assuming classes or a global powerset.

Definitions. For each observer/worldline i, let $\mathcal{A}_i$ be the observer's operational Boolean/σ-algebra (measurable/reportable sets induced by m_i). Let

$$W \;:=\; \bigcup_i \mathcal{A}_i$$

be the intersubjective base. Let S denote the family of *stratified* formulas (per our typing discipline), and write

$$\mathrm{StratComp}(\mathsf{S}; X) \;=\; \big\{\, \{\, u \in U : \varphi(u, \bar{a}) \,\} \;:\; \varphi \in \mathsf{S},\; \bar{a} \in X \,\big\},$$

the stratified comprehensions whose parameters lie in X.

Identity-safe closure (no global powerset). Define an inductive closure over families of subsets of U as follows:

$$\begin{aligned} F_0 &:= W, \\ F_{n+1} &:= F_n \;\cup\; \mathrm{ExtensionalCollapse}(F_n) \;\cup\; \mathrm{StratComp}(\mathsf{S}; F_n) \;\cup\; \mathrm{StdOps}(F_n), \end{aligned}$$

where $\mathrm{StdOps}(F)$ adds the usual set constructions that are definable by stratified formulas with parameters in F (finite pairing, binary union, bounded separation on stratified predicates, Kuratowski pairs/tuples, images under stratified-definable functions). Finally set

$$\mathrm{Cl}(W) \;:=\; \bigcup_{n<\omega} F_n$$

(and allow a standard transfinite continuation if some constructions require it).[2]

Postulates (physics-friendly).

(O) Ontic Coverage (REH hypothesis). For every ontic subset $x \subseteq U$ there exists an observer worldline i and an *in-principle protocol* (possibly realized as a refinement limit along i or an intersubjective refinement chain) such that x is definable by a stratified predicate whose parameters lie in $\mathcal{A}_i$.

(R) Refinement Directedness (intersubjective density). The family $\{\mathcal{A}_i\}_i$ is directed under refinement: for any i, j there exists k (or a refinement chain) with $\mathcal{A}_i \cup \mathcal{A}_j \subseteq \mathrm{Gen}(\mathcal{A}_k)$, where Gen closes under finite joins and the stratified parameters used in (O).

Definition 11 (In-principle protocol)**.** *An in-principle protocol for observer i is a finite specification of a directed family $\{m_{i,\alpha}\}_{\alpha \in D}$ of measurement/refinement procedures together with stratified predicate schemas $\{\varphi_\alpha\}_{\alpha \in D}$ (parameters in $\mathcal{A}_i$) such that the induced sets $S_\alpha := \{\, u \in U : \varphi_\alpha(u) \,\}$ form a refinement net that converges (nested containment or a chosen measurable-set convergence) to a limit set S. The claim "$S = x$" is meaningful only when the protocol is explicitly given.*

[2]This construction is purely *operational*: it never quantifies over a global $\mathcal{P}(U)$; at each stage it only uses sets already present plus stratified comprehension with parameters in the current stage.

Theorem 1 (OC-Universe)**.** *Let $V := \mathrm{Cl}(W)$. Under* (O) *and* (R)*, V exists (by construction) and equals the ontic domain U. Thus a* universal set *is obtained as a consequence of operational closure; no global powerset or class axiom is assumed.*

Proof sketch. Existence: $\mathrm{Cl}(W)$ is the union of the inductive stages, hence a set of subsets of U. ($V \subseteq U$): immediate—every member is built as a subset of U by construction. ($U \subseteq V$): By (O), each ontic set x is definable via a stratified predicate with parameters in some $\mathcal{A}_i \subseteq W$ (possibly in the limit of a refinement net). By closure under stratified comprehension and (R) (which ensures required parameters can be collected operationally), x is added at some stage, hence $x \in V$. □

Remarks and caveats.

- *In-principle vs practice.* (O) is an empirical claim about capacities of physics; it allows refinement/limit constructions but requires protocols to be finitely specifiable. It does not guarantee tractable reconstruction by finite agents or realizability in finite time.
- *Computability and irrationals.* Some limit sets (e.g., exact digit sets) may be definable but not computable to arbitrary precision within finite resources. REH permits such definability while acknowledging practical inaccessibility.
- *Boundary to the imaginary.* A set is "imaginary" for an observer when no explicit in-principle protocol (per the definition) exists that converges to it. Different worldlines may draw different real/imaginary boundaries.

Lemma 1 (Sandboxing)**.** *If φ is unstratified then $\{x : \varphi(x)\}$ does not denote a set; contradictions cannot escape into the ontic domain. They remain as non-denoting conditions within the interior lining.*

Lemma 2 (Refinement Monotonicity)**.** *If $\Pi \preceq \Pi'$ (i.e., Π' refines Π) then $\mathcal{A}_\Pi \subseteq \mathcal{A}_{\Pi'}$ and $\mathrm{Cl}(\mathcal{A}_\Pi) \subseteq \mathrm{Cl}(\mathcal{A}_{\Pi'})$. Refinement cannot conjure contradiction; it only localizes it.*

Remark. This lemma aligns with the experiential arrow later: update/refinement sequences are monotone in information alignment.

2.5 Taxonomy of inference outcomes

Let $\mathcal{I} = (T, R, \varphi)$ be an inference operation and let agent i be the evaluator. Recall:

$$\mathrm{Valid}(T, R, \varphi), \qquad \mathrm{True}_i(\varphi) \text{ (truth under } I_i : L \to \mathcal{B}_i), \qquad \mathrm{PhysRealizable}(\mathcal{I}).$$

For §2.5 we distinguish two interpretations used by an agent when helpful:

$$\mathrm{True}_{\mathrm{phys}}(\varphi) \;:\Longleftrightarrow\; I_{\mathrm{phys}} \models \varphi \quad \text{(operational/measurement interpretation)}, \qquad \mathrm{True}_{\mathrm{ment}}(\varphi) \;:\Longleftrightarrow\; I_{\mathrm{me}}$$

We also use the congruence-warrant notion from §**??**: $\mathrm{Congruence_warrant}_i(\cdot)$.

Cases.

(i) **Neither valid nor sound.**

$$\neg\,\mathrm{Valid}(T, R, \varphi) \;\text{ and }\; \forall j\; \neg\,\mathrm{True}_j(\varphi).$$

Example: malformed proof; inconsistent premises collapse the derivation.

(ii) **Valid, but one or more premises are not sound for some agent.**

$$\mathrm{Valid}(T, R, \varphi) \;\text{ and }\; \exists j\; \exists \psi \in T : \;\neg\, \mathrm{True}_j(\psi).$$

Example: a correct syllogism built from false empirical premises.

(iii) **Warrant-failure (rule misuse).** All premises true for agent i, but a rule in R lacks congruence warrant:

$$\mathrm{Valid}(T, R, \varphi), \quad \forall \psi \in T : \mathrm{True}_i(\psi), \quad \exists r \in R : \;\neg\, \mathrm{Congruence_warrant}_i(r).$$

Example: EQ-Repl applied to a predicate that covertly distinguishes microstates i cannot ignore.

(iv) **Realized Validity (Real/Realization).**

$$\mathrm{Valid}(T, R, \varphi) \;\wedge\; \mathrm{True}_i(\varphi) \;\wedge\; \mathrm{PhysRealizable}(\mathcal{I}).$$

Example: a physics prediction derived and then confirmed experimentally.

(v) **Physically true, conceptually mislabelled (interpretation disparity).**

$$\mathrm{Valid}(T, R, \varphi) \;\wedge\; \mathrm{True}_{\mathrm{phys}}(\varphi) \;\wedge\; \neg\, \mathrm{True}_{\mathrm{ment}}(\varphi).$$

Example: the effect is observed, but the agent's conceptual scheme misidentifies the mechanism.

(vi) **Imaginary / mental-only.**

$$\mathrm{Valid}(T, R, \varphi) \;\wedge\; \mathrm{True}_{\mathrm{ment}}(\varphi) \;\wedge\; \neg\, \mathrm{True}_{\mathrm{phys}}(\varphi).$$

Subcases:

- *Hypothetical:* premises are counterfactual; internally coherent but uninstantiated.
- *Paradoxical:* conflicts with operational definitions; remains sandboxed (non-denoting) for physics.

Compact table (scan view).

Case	Label	Logical conditions
(i)	Neither valid nor sound	$\neg\,\mathrm{Valid}$ and $\forall j\, \neg\, \mathrm{True}_j(\varphi)$
(ii)	Valid, false premise(s)	Valid and $\exists j\, \exists \psi \in T : \neg\, \mathrm{True}_j(\psi)$
(iii)	Warrant-failure	Valid, $\forall \psi \in T\, \mathrm{True}_i(\psi)$, and $\exists r \in R : \neg\, \mathrm{Congruence_warrant}_i(r)$
(iv)	Realized Validity	Valid and $\mathrm{True}_i(\varphi)$ and $\mathrm{PhysRealizable}(\mathcal{I})$
(v)	Physically true, conceptually false	Valid and $\mathrm{True}_{\mathrm{phys}}(\varphi)$ and $\neg\, \mathrm{True}_{\mathrm{ment}}(\varphi)$
(vi)	Imaginary / mental-only	Valid and $\mathrm{True}_{\mathrm{ment}}(\varphi)$ and $\neg\, \mathrm{True}_{\mathrm{phys}}(\varphi)$

The table is schematic: in practice we also track $\mathrm{PhysRealizable}(\mathcal{I})$ and whether externalization succeeds (cf. §**??**).

2.6 Important lemmas (short statements + sketches)

Lemma 3 (Sandboxing)**.** *If set formation is restricted to predicates that are measurable in some $\mathcal{A}_o$ (or identity-safe/stratified), then any contradiction derivable purely syntactically from an agent's internal theory T cannot, by itself, produce a contradiction in any observable algebra $\mathcal{A}_o$ unless the agent* externalizes *the contradictory construction and the externalized predicates are operationally measurable.*

Sketch. Contradictions in T are syntactic artifacts of rules/definitions chosen in the calculus. Observable algebras contain only measurable sets. Under the measurement-level comprehension firewall, the map from syntactic predicates to sets in $\mathcal{A}_o$ is partial: an illicit (unmeasurable or unstratified) comprehension has no denotation in $\mathcal{A}_o$. Hence a purely syntactic contradiction has no image in any $\mathcal{A}_o$. Only when an agent publishes data/procedures and attempts to identify the syntactic construction with an observable set does it become a physical claim subject to test; absent that, the contradiction remains sandboxed. □

Lemma 4 (Substitution warrant ⇒ EQ-Repl sound)**.** *If* $\text{Congruence_warrant}_i(\cdot)$ *holds for all predicates and functions used in R_i, then EQ-Repl (Leibniz substitution) is sound for agent i relative to those predicates/functions.*

Sketch. By the warrant, observationally indistinguishable tokens (equal under i's criteria) preserve the truth of atomic predicates/functions. By compositionality of the satisfaction relation, replacement of equals by equals preserves truth for all formulas built from those atoms using R_i's connectives/quantifiers. Hence every application of EQ-Repl in R_i is truth-preserving under I_i. □

Theorem 2 (Shared interpretation ⇒ intersubjective soundness)**.** *Let agents i, j share an interpretation I (i.e., $I_i = I_j = I$). If $\mathcal{I} = (T, R, \varphi)$ satisfies* $\text{Valid}(T, R, \varphi)$ *and* $\text{PhysRealizable}(\mathcal{I})$*, then $I \models \varphi$ after a successful execution of the physical test.*

Sketch. Validity supplies a derivation of φ from T using R. Physical realizability provides a protocol to adjudicate the premises in T and the conclusion φ on U within tolerances. With $I_i = I_j = I$, denotations of non-logical symbols are shared, so the tested outcomes jointly determine satisfaction under I. A successful realization makes $I \models \varphi$. □

Remark (Pluralism). REH's ontic core is logic-agnostic. We present in classical logic for clarity, but the OC-Universe construction needs only stratified comprehension + directed refinement. Intuitionistic or paraconsistent tracks can be pursued with the same sandboxing discipline.

2.7 Operational rules & editorial commitments

From this point forward we adopt these working constraints (they are this section's filter):

1) **Root Principle (identity).** $\forall u \in U : \; u = u$ holds for the ontic domain U.

2) **Comprehension firewall.** Any comprehension / set-formation used to make *ontic* claims must be *measurement-level* (operational, in some $\mathcal{A}_o$) or explicitly marked *stratified* (cf. Appendix B).

3) **Classical principles are agent-level.** Non-contradiction and excluded middle may be adopted by an agent for internal reasoning; every *externalized* use must carry a semantic warrant (shared interpretation or congruence warrant; §**??**).

4) **Continuum as modelling.** Continuum-based constructions are allowed only with stated measurability/operationalization; otherwise they remain modelling idealizations.

Audit convention. Where helpful, we annotate comprehension instances with badges: [OP] (operational/measurable) and/or [STRAT] (identity-safe/stratified).

Section 2 Conclusion

Section 2 establishes the minimal metaphysical kernel—*identity*—and the operational firewall that prevents logical constructions from leaking contradictions into physics. It separates the ontic level (what exists) from the semantic/epistemic constructions agents use to reason about it, and shows how soundness is tested by reality. From here on we will state explicitly any additional logical axiom or comprehension rule we introduce and record its operational justification (measurement mapping, congruence warrant, or stratification). This keeps the remainder of the manuscript tightly grounded and auditable: unless a principle is named here and motivated, it will not be used to claim physical truths.

3 From Noise to Network (Genesis → Resonance)

Setup and notation

Let η denote an aperiodic "noise substrate" (not assumed metric a priori). We represent local variation by a scalar field σ^2 over an indexable set of sites $\mathcal{X}$ (which may be continuous or discrete).

Definition 12 (Gradient-detectable non-uniformity)**.** *We say η is non-uniform on a region $R \subseteq \mathcal{X}$ if there exists a bounded linear functional $\mathcal{D}$ (difference/derivative operator appropriate to R) such that $(\mathcal{D}\sigma^2)(x) \neq 0$ on a set of nonzero measure.*

Definition 13 (Seed rule and propagation kernel)**.** *Fix (i) a seed selector $\mathsf{seed}(\eta)$ that chooses $x_\star \in \mathcal{X}$ maximizing $|(\mathcal{D}\sigma^2)(x)|$ under a tie-breaking rule, and (ii) a local propagation kernel $K(x \to y) \geq 0$ with $\sum_y K(x \to y) = 1$ on the neighborhood $\mathcal{N}(x)$.*

Definition 14 (Genesis process)**.** *Initialize $G_0 = (V_0, E_0)$ with $V_0 = \{x_\star\}$, $E_0 = \emptyset$. For $t \geq 0$, grow*

$$V_{t+1} = V_t \cup \{y \in \mathcal{N}(x) : K(x \to y) > \tau \text{ for some } x \in V_t\}, \quad E_{t+1} = E_t \cup \{(x, y) : x \in V_t,\ y \in V_{t+1},\ K(x \to$$

for a fixed threshold $\tau \in (0, 1)$. Let $G = \bigcup_{t \geq 0} G_t$.

Lemma: Non-uniform noise ⇒ gradient ⇒ first node/edge

Claim. A non-uniform initial noise field possesses at least one gradient; any distinguished difference seeds an initial node/edge, and iterative propagation generates a network.

Sketch. Let η denote the (aperiodic) noise substrate with local variance $\sigma^2(x)$. If $\nabla\sigma^2 \neq 0$ on a set of nonzero measure, define the first relational edge along the maximal gradient line; propagation by local response yields a graph $G_1 = (V_1, E_1)$, then G_{t+1} by adding edges along induced gradients. (Formalize with thresholded response kernels.)

Lemma 5 (Non-uniform noise seeds a network)**.** *If η is gradient-detectably non-uniform on some region R, then the genesis process produces at least one edge and, under finite connectivity of neighborhoods, a nontrivial network G with $|E| \geq 1$ and $|V| \geq 2$.*

Sketch. Non-uniformity yields $x_\star = \mathsf{seed}(\eta)$ with $|(\mathcal{D}\sigma^2)(x_\star)| > 0$. By continuity or local finiteness, there exists $y \in \mathcal{N}(x_\star)$ with $K(x_\star \to y) > \tau$, hence $(x_\star, y) \in E_1$. Iterating, each frontier vertex has at least one neighbor eligible by the kernel (unless the region is exhausted), so $|V|$ and $|E|$ grow until saturation. $\square$

Lemma: Iterated transformations ⇒ harmonics (resonance)

Claim. Repeated gradient-following with feedback yields scale- and frame-relative harmonics: locally particle-like stability, globally wave-like order.

Sketch. Iterate a diffusion–reaction update on G_t: $x_{t+1} = \Phi(L_t x_t)$ with L_t a graph Laplacian. Fixed/limit cycles ↔ "particles"; spectral bands ↔ "waves". Frame-dependence enters via observer-induced coarse-graining (cf. §2.3).

Lemma 6 (Resonant harmonics from iterative response)**.** *Let $G_t = (V_t, E_t)$ be the network after t genesis steps. Consider the state update*

$$x_{t+1} = \Phi_t\, x_t, \qquad \Phi_t := \Psi\big(L_t\big), \quad L_t := D_t - A_t,$$

where A_t is the weighted adjacency, D_t the degree matrix, L_t the (possibly normalized) Laplacian, and Ψ a Lipschitz nonlinearity (reaction term). Then:

1. *Spectral bands of L_t support distributed, wave-like modes (diffusive/oscillatory patterns).*
2. *Stable/limit-cycle attractors of the nonlinear map Φ_t localize as particle-like "resonance cores."*
3. *Coarse-graining by an observer's partition (§2.3) yields frame-relative stability: the same dynamics may present as particle-like under one partition and wave-like under a finer one.*

Sketch. (1) Linearized dynamics near $x = 0$: $x_{t+1} \approx \Psi'(0)\, e^{-\beta L_t} x_t$ (for suitable $\beta > 0$), so modes evolve along the spectrum of L_t; banded persistence manifests as wave-like order. (2) With nonlinearity, standard results on dissipative graph dynamical systems give existence of attractors; localized fixed points/limit cycles correspond to particle-like cores pinned by topology/weights. (3) Coarse-graining merges high-frequency modes; effective stability becomes partition-dependent (observer frame). $\square$

Algorithm 1: Genesis → Resonance: gradient-seeded network and harmonic emergence

Input: Domain $\mathcal{X}$; noise field $\sigma^2 : \mathcal{X} \to \mathbb{R}_{\geq 0}$; difference operator $\mathcal{D}$; neighborhood rule $\mathcal{N}(\cdot)$; threshold $\tau \in (0,1)$; kernel K; dynamics nonlinearity Ψ; steps $(T_{\text{grow}}, T_{\text{dyn}})$

Output: Graph sequence $\{G_t\}$; states $\{x_t\}$; spectral logs; attractor summaries

$x_\star \leftarrow \arg\max_{x \in \mathcal{X}} |(\mathcal{D}\sigma^2)(x)|$ `// seed on strongest gradient`
$G_0 \leftarrow (V_0 = \{x_\star\}, E_0 = \emptyset)$
for $t \leftarrow 0$ **to** $T_{\text{grow}} - 1$ **do**
  foreach $x \in V_t$ **do**
    foreach $y \in \mathcal{N}(x)$ **do**
      if $K(x \to y) > \tau$ **then**
        $V_{t+1} \leftarrow V_t \cup \{y\}$; $E_{t+1} \leftarrow E_t \cup \{(x,y)\}$; $w_{xy} \leftarrow K(x \to y)$
$G \leftarrow G_{T_{\text{grow}}}$; build $A, D, L = D - A$; optionally normalized $\mathcal{L}$
$x_0 \leftarrow$ small random field on V (or stimulus at seed/frontier)
for $t \leftarrow 0$ **to** $T_{\text{dyn}} - 1$ **do**
  $x_{t+1} \leftarrow \Psi(L\, x_t)$ `// e.g.,` $\Psi(z) = z - \alpha z^3$ `or` $\Psi(z) = \tanh(\beta z)$
  if *topology_feedback* **then**
    foreach $(u,v) \in E$ **do**
      $w_{uv} \leftarrow \text{clip}\big(w_{uv} + \lambda\, |x_t(u) - x_t(v)|,\ w_{\min}, w_{\max}\big)$
    update A, D, L
  if $t \bmod k_{\log} = 0$ **then**
    compute leading m eigenpairs (λ_j, ϕ_j) of L (power/Lanczos)
    energy in bands: $E_j(t) = \langle x_t, \phi_j \rangle^2$
    localization (IPR): $\text{IPR}(x_t) = \sum_u x_t(u)^4 / \|x_t\|_2^4$
    mix time proxy: $\text{Tr}(e^{-sL})$ or $\|P^s - \pi\|$ for small s
    save $\{E_j(t)\}$, IPR, norms, and graph stats (degree, clustering)
  if $t \geq h$ ***and*** $\|x_t - x_{t-h}\|_2 < \varepsilon$ **then**
    record attractor (type $\in$ {fixed,cycle}, support, period $\leq h$)
return $\{G_t\}, \{x_t\}$, spectral/attractor logs

Tools: geodesics, diffusion, and spectra on G

Geodesics are shortest-paths under edge weights w_e. Diffusion uses a row-stochastic $P = D^{-1}A$ with $p_{t+1} = Pp_t$. Spectral content uses eigenpairs (λ_k, ϕ_k) of L; heat kernels $H_t = e^{-tL}$ and wave kernels $\cos(t\sqrt{L})$ organize resonance structure.

4 Spacetime as Emergent Topology

Theorem 3 (Reductio: spacetime is not fundamental)**.** *If smooth spacetime (M, g) is taken as the fundamental ontic substrate, then (i) generic evolution admits curvature/energy divergences (singularities), and (ii) reconciliation with quantum field regularization requires non-ontic cutoffs. Therefore, spacetime must be an* emergent visualization *of a deeper relational topology/density.*

Sketch. (i) Under classical hypotheses (energy conditions, global hyperbolicity), geodesic incompleteness follows for broad classes of matter/curvature profiles (singularity behavior).

(ii) Interacting QFTs on (M, g) require renormalization; ultraviolet cutoffs are non-ontic if M is fundamental. Replacing the substrate by a finite/intrinsically discrete relational object $G = (V, E, w)$ bounds curvature via combinatorial quantities and removes the need for external cutoffs; the observed continuum then arises as a scaling limit (§10). □

Graph geometry toolbox

Let $G = (V, E, w)$ be a weighted graph with symmetric weights $w_{uv} \geq 0$.

$$A_{uv} := w_{uv}, \qquad d_u := \sum_v A_{uv}, \qquad D := \mathrm{diag}(d_u), \tag{1}$$

$$L := D - A \quad \text{(combinatorial Laplacian)}, \qquad \mathcal{L} := I - D^{-1/2} A D^{-1/2} \quad \text{(normalized)}. \tag{2}$$

Geodesics are shortest paths w.r.t. edge lengths $\ell_{uv} := 1/w_{uv}$ (or another positive length). Diffusion uses a Markov kernel $P := D^{-1}A$ with $p_{t+1} = P\, p_t$ (an *effective* time parameter).

Discrete Ricci analogs. For $u \sim v$, let μ_u be a one-step neighborhood measure (e.g., $\mu_u(v) = A_{uv}/d_u$). Ollivier's curvature:

$$\kappa_{\mathrm{Oll}}(u, v) \;:=\; 1 - \frac{\mathcal{W}_1(\mu_u, \mu_v)}{\ell_{uv}}, \tag{3}$$

where $\mathcal{W}_1$ is the 1-Wasserstein distance under edge lengths. Forman's curvature on an edge $e = (u, v)$ with weight w_e:

$$\kappa_{\mathrm{Forman}}(e) \;:=\; \frac{w_{uv}}{w_{uv}}\left(\frac{w_{uv}}{w_{uv}}\right) + \sum_{\substack{e'=(u,\cdot)\\ e'\neq e}} \frac{w_{uv}}{\sqrt{w_{uv} w_{e'}}} + \sum_{\substack{e''=(v,\cdot)\\ e''\neq e}} \frac{w_{uv}}{\sqrt{w_{uv} w_{e''}}} - \sum_{\square \ni e} \frac{w_{uv}}{\sqrt{w_{uv} w_{\square}}}, \tag{4}$$

with the last sum ranging over 2-cells (if present) or set to 0 for simple graphs; choose a standard variant consistently throughout.

Remark (Info redshift and focusing). Negative edge curvature (by (3) or (4)) implies geodesic focusing and slower diffusion mixing; we interpret this as *gravitational* information redshift along weighted paths. Positive curvature defocuses and accelerates mixing.

Proposition 2 (Continuum calibration (discrete → GR))**.** *Let G_ϵ be a random geometric (or Vietoris–Rips) graph embedded in a smooth Riemannian manifold (M, g) at scale $\epsilon \to 0$, with weights chosen from a smooth kernel k_ϵ and edge lengths $\ell_{uv} = \mathrm{dist}_g(u, v)$. Then, after suitable normalization,*

$$\kappa^{(\epsilon)}_{\mathrm{Oll}}(u, v) \;\underset{\epsilon\to 0}{\longrightarrow}\; c_1\, \mathrm{Ric}_g(\dot\gamma, \dot\gamma) + O(\epsilon), \qquad \mathcal{L}_\epsilon \;\underset{\epsilon\to 0}{\longrightarrow}\; c_2\, \Delta_g,$$

in the sense of measured Gromov–Hausdorff convergence, for constants $c_1, c_2 > 0$ determined by k_ϵ.

Sketch. Known Laplacian consistency results give $\mathcal{L}_\epsilon \to c_2 \Delta_g$. For transport-based curvatures, first-order expansions of $\mathcal{W}_1(\mu_u, \mu_v)$ over small geodesic steps relate to sectional/Ricci components along the edge direction $\dot\gamma$; normalization yields the limit. Details depend on kernel/graph model and boundary conditions. □

Remark (Effective Einstein correspondence). Under coarse-graining, discrete curvature scalars aggregated over edges/faces can be regressed onto continuum invariants (e.g., Ric, R, or the Einstein tensor $G_{\mu\nu}$). Section 10 proposes a calibration pipeline (lensing/time-dilation benchmarks) to empirically fit c_1, c_2 and assess residuals.

Symmetry → conservation on networks

Definition 15 (Network symmetry and gauge)**.** *An automorphism $\pi \in \mathrm{Aut}(G)$ is a vertex relabeling that preserves A (and w). A* gauge transformation *is an equivalence of labelings that leaves the family of admissible observables and dynamics invariant (e.g., $x \mapsto Ux$ with U block-orthogonal on isotypic components).*

Proposition 3 (Noether-like correspondence on G)**.** *If the dynamics $x_{t+1} = F(A, D; x_t)$ is invariant under a symmetry group $\mathcal{G} \subseteq \mathrm{Aut}(G)$ (i.e., $F(A, D; Ux) = UF(A, D; x)$ for all $U \in \mathcal{G}$), then there exists a family of conserved quantities $C_U(x) := \langle x, Ux\rangle$ (or their appropriate nonlinear generalizations) constant along trajectories. For continuous symmetries (e.g., phase rotations per connected component), conserved currents arise from the corresponding infinitesimal generators.*

Sketch. Invariance implies $C_U(x_{t+1}) = \langle F(\cdot; x_t), UF(\cdot; x_t)\rangle = \langle U^{-1}F(\cdot; Ux_t), UF(\cdot; x_t)\rangle = C_U(x_t)$ for unitary/orthogonal U; extend by density for admissible nonlinearities. □

Remark (Flat connections). On cycle spaces, gauge potentials with trivial holonomy (product of edge phases = 1 around any cycle) define *flat* sectors. Nontrivial holonomy encodes frustration/defects and shifts spectra (Aharonov–Bohm–like effects) that mimic "fields" without introducing a background manifold.

5 Quanta: Parsing Creates Discreteness

Manifold as effective limit

Continuous descriptions approximate dense relational regions; the "continuum" is a scaling limit of G under refinement (cf. Prop. 2).

Proposition 4 (Continuum as refinement limit)**.** *Let $\{G_\epsilon\}$ be refinement graphs sampled from (M, g) as in Prop. 2. For Laplacians $\mathcal{L}_\epsilon$ normalized compatibly with sampling, $\mathcal{L}_\epsilon \to c_2\,\Delta_g$ in operator sense as $\epsilon \to 0$, and band-limited eigenmodes of $\mathcal{L}_\epsilon$ converge (weakly) to those of Δ_g.*

Sketch. Standard graph-to-manifold Laplacian consistency under kernel sampling and density correction; spectral convergence follows from compactness and perturbation results (as in §4). □

Collapse (operational)

Selection of a resonance branch consistent with an observer's lining (§2.2); mathematically, a POVM on G.

Definition 16 (POVM on a graph state)**.** *Let $x \in \mathbb{R}^V$ (or $\mathbb{C}^V$) be a unit-norm state on vertices. A POVM is a finite family $\mathsf{E} = \{E_k\}_{k=1}^m$ of positive semidefinite operators on $\mathbb{R}^V$ such that $\sum_k E_k = I$. Outcome k occurs with probability $p_k = \langle x, E_k x\rangle$ and post-measurement state $x' \propto E_k^{1/2} x$.*

Proposition 5 (Operational collapse on G)**.** *If outcome k is realized, the network undergoes a connectivity update $G \mapsto G^{(k)}$ by a rule*

$$w_{uv}^{(k)} \;=\; \Phi_k\big(w_{uv},\, |x(u) - x(v)|\big), \qquad \textit{with} \quad \Phi_k \textit{ monotone in } |x(u) - x(v)|.$$

Equivalently, the Laplacian updates to $L^{(k)} = D^{(k)} - A^{(k)}$ with $A^{(k)} = (w_{uv}^{(k)})$.

Concrete connectivity updates (choices). *Given a post-measurement state x' after outcome k, define the updated weights by an edgewise contrast $c_{uv} := |x'(u) - x'(v)|$.*

1. Exponential down-weighting (contrast-sensitive diffusion / Perona–Malik style):

$$w_{uv}^{(k)} \;:=\; w_{uv}\,\exp\big(-\beta_k\, c_{uv}^2\big), \qquad \beta_k > 0.$$

2. Logistic gate (soft threshold θ_k):

$$w_{uv}^{(k)} \;:=\; \frac{w_{uv}}{1+\exp\big(\beta_k(c_{uv}-\theta_k)\big)}, \qquad \beta_k > 0,\ \theta_k \geq 0.$$

3. Hard prune + renormalize (sparsifying parse):

$$w_{uv}^{(k)} \;:=\; \begin{cases} w_{uv}, & c_{uv} \leq \theta_k, \\ 0, & c_{uv} > \theta_k, \end{cases} \quad \textit{then} \quad \tilde{w}_{uv}^{(k)} \;:=\; \frac{w_{uv}^{(k)}}{\sum_{v'} w_{uv'}^{(k)}} \sum_{v'} w_{uv'},$$

which preserves each vertex's total incident weight.

These rules preserve symmetry (since $c_{uv} = c_{vu}$) and nonnegativity; the degree and Laplacian update as $d_u^{(k)} = \sum_v w_{uv}^{(k)}$, $L^{(k)} = D^{(k)} - A^{(k)}$ with $A^{(k)} = (w_{uv}^{(k)})$. Choice (1) yields smooth red/blue-shift of edges; (2) gives soft band-gaps; (3) implements measurement-as-prune while keeping node strength.

Lemma 7 (PSD and contractivity)**.** *If L is PSD and $A^{(k)} = A \circ G_k$ with $G_k(u,v) \in [0,1]$ symmetric (Hadamard product), then $L^{(k)}$ is PSD. Moreover, for any x, $x^\top L^{(k)} x \leq x^\top L x$.*

Sketch. For any x, $x^\top L x = \frac{1}{2}\sum_{u,v} w_{uv}\,(x_u - x_v)^2$. Since $0 \leq G_k \leq 1$, replace w_{uv} by $w_{uv} G_k(u,v)$ to get a smaller (or equal) nonnegative sum. PSD follows. □

Sketch. The POVM outcome defines which features of x become operationally reportable; edges inconsistent with the reported contrast are down-weighted or contracted by a monotone rule (measurement-as-parse). This implements the observer's admissible partition as an operator-induced reweighting. □

Remark (Born rule and Naimark). $p_k = \langle x, E_k x\rangle$ is the operational frequency. Any POVM on V admits a Naimark dilation to a projective measurement on $V \cup V_{\text{anc}}$, i.e., an enlarged graph with ancilla nodes; thus "collapse" can be represented as projection plus restriction.

"Abracadabra": parsing is topologically transformative

Finite-resolution parsing discretizes and *changes connectivity.*

Definition 17 (Parsing operator and quotient graph)**.** *A parsing operator is a surjection $\pi : V \to V'$ that induces a partition $\mathcal{P} = \{C_1, \ldots, C_{|V'|}\}$ of V. The quotient (or coarse-grained) graph $G' = (V', E', w')$ has weights*

$$w'_{ab} \;:=\; \sum_{\substack{u \in C_a \\ v \in C_b}} w_{uv}, \qquad a \neq b, \quad \textit{and} \quad w'_{aa} := 0.$$

Let $C \in \{0,1\}^{|V'|\times|V|}$ be the cluster-incidence matrix $C_{a,u} = 1$ iff $u \in C_a$, and $S = \operatorname{diag}(|C_1|, \ldots, |C_{|V'|}|)$. Define the averaging map $\Pi_{\text{parse}} := S^{-1}C$ and the coarse Laplacian

$$L' \;:=\; \Pi_{\text{parse}}\, L\, \Pi_{\text{parse}}^\top.$$

Theorem 4 (Parsing transforms topology and spectra)**.** *For any nontrivial partition $\mathcal{P}$,*

1. *$G \mapsto G'$ alters adjacency and geodesics in general (changes in cut structure; possible edge contractions).*

2. *The Laplacian spectrum obeys interlacing: if $0 = \lambda_1 \leq \cdots \leq \lambda_n$ are eigenvalues of L and $0 = \lambda'_1 \leq \cdots \leq \lambda'_{n'}$ those of L', then*

$$\lambda_i \;\leq\; \lambda'_i \;\leq\; \lambda_{i+|V|-|V'|} \qquad (1 \leq i \leq |V'|).$$

3. *Block-constant modes on $\mathcal{P}$ descend exactly: if x is constant on each C_a, then $L'x' = (\Pi_{\text{parse}} L \Pi_{\text{parse}}^\top)x' = \Pi_{\text{parse}}(Lx)$, so eigenpairs restricted to the block-constant subspace are preserved.*

Sketch. (1) Follows from quotient construction: merging vertices modifies edge multiplicities and shortest paths. (2) Interlacing is a standard consequence of Courant–Fischer with compression by Π_{parse}. (3) Direct calculation on the block-constant subspace. □

Corollary 1 (Emergent discreteness without discrete priors)**.** *Any finite-resolution parse π yields a finite-dimensional operator L' with a* discrete *spectrum, and measurement-driven reweightings $L \mapsto L^{(k)}$ followed by parse π produce piecewise-constant "eigentextures" on V' even if the pre-parse description of (M, g) is continuous.*

Remark (Two routes to the same discreteness). (i) *Coarse-graining*: Π_{parse} compresses L (spectral interlacing). (ii) *State-conditioned update*: POVM outcome k changes $A \to A^{(k)}$ (non-commuting with L in general), after which coarse-graining yields new bands and gaps. Both routes give discrete spectra without postulating discreteness ab initio.

6 Dual Causality, CPT, and Arrows

Micro (time symmetry)

CPT symmetry; Stueckelberg–Feynman reinterpretation (electron backward-in-time $\leftrightarrow$ positron forward); propagators as advanced + retarded; microcausality preserved.

Emitter/Detector Dualism

Cameras/screens as adjoints; perception as retrodiction via a transpose operator $T^\top$ acting on latent causes.

Arrows

- **Thermodynamic arrow:** low-entropy boundary selects retarded behavior macroscopically.

- **Logical arrow:** interior linings well-order experience by compressing contradiction; physics remains locally time-symmetric.

7 The Computational Lens

Turing-embeddability

The universe permits universal computation; incompleteness limits *formal systems*, not reality.

Logic as complement to physics

Computation is physical $\Rightarrow$ universal computation must be permitted somewhere in U; substrates differ, computability class persists.

Incompleteness (practical)

No device solves Halting on all inputs; bounded subsystems approximate decisions on dense subsets (PAC-style bounds, finite-time truncations).

Paradox compression/diffusion

Contradiction is localized in linings (sandboxing, §2.6); communication diffuses inconsistency without global collapse.

8 Thermodynamics and the Logical Arrow

Entropy as relational multiplicity

Use von Neumann entropy on diffusion/Laplacian spectra or Markov kernels:

$$S(\rho) = -\mathrm{Tr}(\rho \log \rho), \quad \rho \propto e^{-\beta L} \ \text{ or } \ P.$$

Arrow = growth of accessible microstates under gradients with low-entropy boundary.

Experiential time

Bayesian/variational updates form a supermartingale in divergence to data-implied models; the "felt" arrow = order of successful parses.

9 Relativity, Frames, and Intersubjectivity

Tautological solipsism

For each observer: "I am the only mind" (others appear as brains); no contradiction across frames under §2.2.

Resonance at infinity

Parallel worldlines never intersect yet "meet" via synchronized emergence; art/language/culture act as convergence engines.

10 Empirical Signatures and Tests

Graph–GR calibration

Recover lensing/time dilation from discrete curvature; calibrate Ollivier/Forman curvature to Einstein tensor in the continuum limit.

Calibration protocol: discrete curvature → GR observables

Goal. Empirically fit the discrete network quantities (Ollivier/Forman curvature, Laplacians, diffusion times) to continuum GR invariants and observables (Ricci/Einst. tensors, lensing, redshift, time dilation), estimating the normalization constants (c_1, c_2) and checking scale stability (§4, Prop. 2).

Inputs. (1) A family of weighted graphs $G_\epsilon = (V, E, w_\epsilon)$ generated from a known spacetime (M, g) by sampling points and connecting kNN/ϵ-radius with lengths $\ell_{uv} \approx \mathrm{dist}_g(u, v)$ and weights $w_{uv} = k_\epsilon(\ell_{uv})$. (2) Ground-truth continuum tensors on sampled points: Ric_g, R_g, $G_{\mu\nu}$, geodesic distances and null geodesics.
Graph features. Compute $A, D, L, \mathcal{L}$, random-walk $P = D^{-1}A$, one-step measures μ_u, $\kappa_{\mathrm{Oll}}(u, v)$, $\kappa_{\mathrm{Forman}}(u, v)$, and diffusion times τ_{mix} or heat kernel traces $\mathrm{Tr}(e^{-tL})$.
Targets. Directional Ricci along edge tangents, R_g, and Einstein components projected to local frames: $\widehat{\mathrm{Ric}}(u, v) := \mathrm{Ric}_g(\dot\gamma_{uv}, \dot\gamma_{uv})$, $\widehat{G}(u) := n^\mu n^\nu G_{\mu\nu}$ for chosen unit timelike n, plus observables: lensing deflections $\hat\alpha$ and gravitational time dilation $\Delta t/t$.
Maps to learn (calibration).

$$\begin{aligned} &\text{(C1)} \quad \kappa_{\mathrm{Oll}}(u, v) \;\mapsto\; c_1\, \widehat{\mathrm{Ric}}(u, v), \qquad \text{(C2)} \quad \mathcal{L} \;\mapsto\; c_2\, \Delta_g, \\ &\text{(C3)} \quad f_\theta(\kappa_{\mathrm{Forman}},\ \mathrm{deg},\ w,\ \mathrm{cycles}) \;\mapsto\; \widehat{R},\ \widehat{G}, \end{aligned}$$

where f_θ is a low-capacity regressor (linear or shallow polynomial) to avoid overfitting.
Loss. Mean-squared errors with scale regularization and multi-scale consistency:

$$\mathcal{L}_{\mathrm{curv}} = \sum_{(u,v)\in E} \big(\kappa_{\mathrm{Oll}}(u, v) - c_1\, \widehat{\mathrm{Ric}}(u, v)\big)^2 + \lambda \|\nabla_\epsilon c_1\|^2, \quad \mathcal{L}_{\mathrm{Lap}} = \big\|\mathcal{L}_\epsilon - c_2\, \Pi\, \Delta_g\, \Pi^\top\big\|_F^2,$$

plus observable losses (lensing, redshift/time dilation) computed from shortest paths/diffusion:

$$\mathcal{L}_{\mathrm{obs}} = \mathrm{MSE}\big(\hat\alpha_{\mathrm{graph}},\ \hat\alpha_g\big) + \mathrm{MSE}\Big(\frac{\Delta t_{\mathrm{mix}}}{t},\ \frac{\Delta t_g}{t}\Big).$$

Total: $\mathcal{L} = \mathcal{L}_{\mathrm{curv}} + \mathcal{L}_{\mathrm{Lap}} + \beta\, \mathcal{L}_{\mathrm{obs}}$.
Validation. Hold out unseen geometries and scales: constant-curvature spaces (S^n, H^n), FLRW slices, Schwarzschild/Kerr equatorial plane, and noisy embeddings. Check: (i) stability of (c_1, c_2) across ϵ, k, (ii) error vs. sampling density, (iii) transfer to perturbed metrics.
Uncertainty. Bootstrap resample vertices/edges; report 95% CIs for (c_1, c_2) and observable residuals. Ablate features (remove cycles/weights) to quantify necessity.
Pass/fail gates. G1: $\mathrm{MAE}(\kappa \to \widehat{\mathrm{Ric}}) < \tau_1$ across scales. G2: $\|\mathcal{L}_\epsilon - c_2\Delta_g\|_F / \|\Delta_g\|_F < \tau_2$. G3: Lensing/time-dilation relative error $< \tau_3$ on held-out metrics.

Datasets (suggested). Synthetic manifolds: constant-curvature (S^2, H^2), FLRW ($a(t)$ tables), Schwarzschild (exact geodesics), Kerr (equatorial), and perturbed metrics (Gaussian bumps). Observables: ray-traced lensing maps, redshift/time-dilation profiles, and geodesic-distance matrices. Realish: weak-lensing mass maps (for qualitative checks) and cosmology mocks consistent with ΛCDM.

Reporting. Provide (c_1, c_2) with CIs, error–vs–scale curves, parity plots (κ vs. $\widehat{\mathrm{Ric}}$), and ablation tables. Release code to regenerate G_ϵ, compute features, and reproduce fits.

Algorithm 2: Calibration pipeline: discrete curvature → GR observables

Input: Manifold (M, g) with oracle for dist_g, tensors $(\mathrm{Ric}_g, R_g, G_{\mu\nu})$ on samples; scales $\{\epsilon\}$; sampler Sample; kernel family $\{k_\epsilon\}$; tolerances (τ_1, τ_2, τ_3)

Output: Normalization (c_1, c_2) with CIs; fitted map f_θ; error reports and pass/fail gates

foreach $\epsilon \in \{\epsilon\}$ **do**
- $X_\epsilon \leftarrow \mathsf{Sample}(M, \epsilon)$ `// point cloud`
- Build graph $G_\epsilon = (V, E, w_\epsilon)$ via kNN or ϵ-radius, edge lengths $\ell_{uv} \approx \mathrm{dist}_g(u, v)$, weights $w_{uv} = k_\epsilon(\ell_{uv})$
- Compute $A, D, L, \mathcal{L}$, random-walk $P = D^{-1}A$
- **foreach** $(u, v) \in E$ **do**
  - $\mu_u, \mu_v \leftarrow$ one-step neighborhood measures
  - $\kappa_{\mathrm{Oll}}(u, v) \leftarrow 1 - \mathcal{W}_1(\mu_u, \mu_v)/\ell_{uv}$
  - $\kappa_{\mathrm{Forman}}(u, v) \leftarrow$ chosen Forman variant
  - $\widehat{\mathrm{Ric}}(u, v) \leftarrow \mathrm{Ric}_g(\dot\gamma_{uv}, \dot\gamma_{uv})$
- Fit c_1 by minimizing $\mathcal{L}_{\mathrm{curv}} = \sum_{(u,v)} \left(\kappa_{\mathrm{Oll}} - c_1 \widehat{\mathrm{Ric}}\right)^2 + \lambda \|\nabla_\epsilon c_1\|^2$
- Fit c_2 by minimizing $\mathcal{L}_{\mathrm{Lap}} = \|\mathcal{L}_\epsilon - c_2 \Pi \Delta_g \Pi^\top\|_F^2$
- Train a low-capacity regressor f_θ to map graph features → $(\widehat{R}, \widehat{G})$ with validation on held-out vertices
- Compute observable errors: lensing $\mathrm{MSE}(\hat\alpha_{\mathrm{graph}}, \hat\alpha_g)$; time dilation $\mathrm{MSE}(\Delta t_{\mathrm{mix}}/t, \Delta t_g/t)$

Aggregate across scales; bootstrap resample vertices/edges to form CIs for (c_1, c_2) and residuals

Evaluate gates:
1. G1: $\mathrm{MAE}(\kappa \to \widehat{\mathrm{Ric}}) < \tau_1$,
2. G2: $\|\mathcal{L}_\epsilon - c_2 \Delta_g\|_F / \|\Delta_g\|_F < \tau_2$,
3. G3: observable relative errors $< \tau_3$ on held-out metrics.

return (c_1, c_2), f_θ, reports

Cosmology

CMB anisotropies as imprints of primordial gradients; predict small deviations from ΛCDM in large-scale graph curvature.

Quantum

Bell-type violations as non-local edges; area laws from boundary cuts; collapse statistics as parse-induced topology changes.

Simulations

Agent-based + spectral-graph demos: (i) resonance from gradients, (ii) geodesic focusing = gravity, (iii) entropy-growth arrow, (iv) adjoint reconstruction = perceptual retrodiction.

11 Discussion and Limitations

Reciprocity assumptions; nonlinear/gain extensions; graph-curvature calibration; phenomenology vs. reduction across registers; genesis as logical inevitability but priors constrained by data.

12 Conclusion

REH unifies emergence in physics and cognition: spacetime from networks; mind as boundary; contradiction localized. Logically taut, operationally testable, mathematically extensible. Next: derive discrete→continuum curvature links, formalize measurement-as-parse via POVMs on graphs, and run end-to-end simulations tying cosmology, quantum signatures, and perceptual adjoints.

A Appendix I — Formal proofs and lemmas (expanded natural-deduction)

Purpose. This appendix supplies fully explicit, auditable proofs for the illustrative arithmetic equalities and structured proofs (or detailed proof sketches) of the lemmas in Section 2.6. We present natural-deduction (ND) derivations with every axiom/rule invocation identified.

Notation / conventions (ND style).

- *Axiom* marks an axiom instance.
- $\forall E$ / $\forall I$ = universal elimination/introduction.
- $\to I$ / $\to E$ = implication introduction/elimination (modus ponens).
- $=\mathrm{Repl}(\varphi)$ indicates an application of *Leibniz substitution* for a subformula/term φ.
- **EQ-Refl** = $\forall x\,(x = x)$; **EQ-Repl** = substitution schema for $=$ (Leibniz).
- **Def-1** = $1S(0)$ used as the axiom $1 = S(0)$.
- **PA3** / **PA4** = Peano addition axioms: $\forall x\,(x + 0 = x)$ and $\forall x\,\forall y\,(x + S(y) = S(x + y))$.

A. Preliminary: equality calculus from EQ-Refl + EQ-Repl

Proposition 6 (Symmetry)**.** $a = b \to b = a$.

ND derivation. 1. $\forall x\,(x = x)$ (Axiom: EQ-Refl)

2. Assume $a = b$ (Assumption)
3. Let $\varphi(x)(x = a)$ (Instance choice)
4. From EQ-Repl on φ: $a = b \to \big(a = a \leftrightarrow b = a\big)$ (= Repl(φ))
5. From 1 by $\forall E$ with $x := a$: $a = a$ ($\forall E$)
6. From 2 and 4 by $\to E$: $a = a \leftrightarrow b = a$ ($\to E$)
7. From 5 and 6: $b = a$ (biconditional elimination)
8. Therefore $a = b \to b = a$ ($\to I$)

□

Proposition 7 (Transitivity)**.** $(a = b \wedge b = c) \to a = c$.

ND derivation. 1. Assume $a = b$ and $b = c$ (Assumptions)

2. Let $\varphi(x)(x = c)$ (Instance choice)
3. From EQ-Repl on φ: $a = b \to \big(a = c \leftrightarrow b = c\big)$ (= Repl(φ))
4. From 1 (second conjunct): $b = c$ (Conjunction elimination)
5. From 1 (first conjunct) and 3 by $\to E$: $a = c \leftrightarrow b = c$ ($\to E$)
6. From 4 and 5: $a = c$ (biconditional elimination)
7. Therefore $(a = b \wedge b = c) \to a = c$ ($\to I$)

□

B. Fully explicit arithmetic ND proofs

Proof A.1 — $1 = 1$ **(trivial instance).**

1. $\forall x\,(x = x)$ (Axiom: EQ-Refl)
2. $1 = 1$ (from 1 by $\forall E$ with $x := 1$)

□

Proof A.2 — $0 + 1 = 1 + 0$ **(detailed ND).** Axioms: EQ-Refl, EQ-Repl, **PA3**: $\forall x\,(x + 0 = x)$, **PA4**: $\forall x\,\forall y\,(x + S(y) = S(x + y))$, **Def-1**: $1 = S(0)$.

1. $1 = S(0)$ (Axiom: Def-1)
2. $\forall x\,(x + 0 = x)$ (Axiom: PA3)
3. $1 + 0 = 1$ (from 2 by $\forall E$ with $x := 1$)
4. $0 + 0 = 0$ (from 2 by $\forall E$ with $x := 0$)
5. $\forall x\,\forall y\,(x + S(y) = S(x + y))$ (Axiom: PA4)
6. $0 + S(0) = S(0 + 0)$ (from 5 by $\forall E$ with $x := 0, y := 0$)
7. From 4 and 6: $0 + S(0) = S(0)$ (= Repl, replace $0 + 0$ by 0)
8. From 1 and prop:symmetry-nd: $S(0) = 1$ (Symmetry)
9. From 7 and 8: $0 + 1 = 1$ (= Repl)
10. $1 + 0 = 1$ (from 3)
11. From 9: $0 + 1 = 1$; from 10 and symmetry: $1 = 1 + 0$; by transitivity: $0 + 1 = 1 + 0$ (prop:transitivity-nd + symmetry)

□

Proof A.3 — $1 = \frac{1}{2} + \frac{1}{2}$ **(definitional).** Introduce a constant h with axiom $h + h = 1$.

1. $h + h = 1$ (Axiom: Def-$\frac{1}{2}$)
2. $1 = h + h$ (Symmetry of equality)
3. Define $\frac{1}{2}h$. Then $1 = \frac{1}{2} + \frac{1}{2}$ by notational substitution.

□

C. Formal proofs / structured proofs of lemmas and theorem

For the lemmas and theorem we provide rigorous structured proofs suitable for formalization. Where fully formal line-by-line ND would be extremely lengthy (induction on formula structure), we give detailed, explicit sketches with clear induction principles stated.

Lemma 8 (Sandboxing; expands Lemma 3)**.** *If comprehension in a theory T is restricted to measurable predicates in some observable algebra $\mathcal{A}_o$ (or to identity-safe/stratified formulas), then from $T \vdash (\psi \wedge \neg\psi)$ it does not follow that a contradiction exists in any observable algebra $\mathcal{A}_o$ unless the agent externalizes the construction and the predicate defining ψ is measurable in $\mathcal{A}_o$.*

Formal proof sketch. (1) Let Comp_T be the comprehension instances admitted by T; by hypothesis, each uses a measurable (or stratified) $\varphi(x)$. The comprehension operator $x \mapsto \{u \in U \mid \varphi(u)\}$ is only defined for such φ.
(2) A syntactic contradiction $\psi \wedge \neg\psi$ arises from rules/definitions; unless at least one comprehension instance used to build ψ (or $\neg\psi$) maps to a measurable predicate, there is no set in $\mathcal{A}_o$ denoting that object.
(3) Therefore no pair of measurable sets in $\mathcal{A}_o$ is produced that simultaneously assert S and its complement as observed; no direct contradiction in $\mathcal{A}_o$ follows.
(4) Only upon *externalization* (publishing protocol/data identifying the syntactic object with an observable set) does the mapping exist; if measurable, the contradiction becomes testable and may conflict with $\mathcal{A}_o$. □

Lemma 9 (Substitution warrant ⇒ EQ-Repl sound; expands Lemma 4)**.** *Suppose for agent i every atomic predicate and function symbol used in R_i satisfies* Congruence_warrant$_i$. *Then EQ-Repl (Leibniz substitution) is sound for agent i relative to those predicates/functions.*

Structured proof by structural induction. **Base (atomic).** For atomic $P(t_1, \ldots, t_n)$, if $t_k = s_k$ (and equalities for other replaced positions), Congruence_warrant$_i(P)$ ensures preservation of truth under I_i when replacing t_k by s_k.
Step (connectives/quantifiers). Assume preservation up to complexity $\leq n$. For $\neg, \wedge, \vee, \rightarrow$, compositional semantics plus IH yield preservation. For $\forall, \exists$, rename bound variables as needed and apply IH to the matrix. Hence EQ-Repl is sound for all formulas built from admissible atoms. □

Theorem 5 (Shared interpretation ⇒ intersubjective soundness; expands Theorem 2)**.** *If agents i, j share interpretation I ($I_i = I_j = I$) and $\mathcal{I} = (T, R, \varphi)$ satisfies* $\mathrm{Valid}(T, R, \varphi)$ *and* $\mathrm{PhysRealizable}(\mathcal{I})$*, then after successful execution of the physical test we have $I \models \varphi$.*

Proof sketch. Validity gives a derivation of φ from T using R. Physical realizability provides a protocol adjudicating premises in T and the observables corresponding to φ on U (within tolerances). With $I_i = I_j = I$, denotations are shared; observed outcomes determine satisfaction under I. A successful realization yields $I \models \varphi$. □

Lemma 10 (Refinement Monotonicity; expands Lemma 2)**.** *If $B_1 \subseteq B_2$ then* $\mathrm{Cl}(B_1) \subseteq \mathrm{Cl}(B_2)$*. In particular, if observers refine their algebras, the resulting OC-Universe grows monotonically (no contradictions are conjured by refinement alone).*

Sketch. By definition, Cl is the least family closed under extensional collapse, stratified comprehension with parameters from the base, and the standard stratified-definable operations. Each clause is monotone in the base; thus inclusion is preserved at every stage of the inductive construction, hence at the limit. □

Editorial notes: EQ-Repl and Anti-Foundation (AFA).

- *EQ-Repl.* Kept as a *rule* in the classical track. It is only used where Congruence_warrant holds; alternate calculi (paraconsistent, intuitionistic) are admissible as parallel tracks.
- *AFA / non-well-founded models.* Allowed as a modelling tool for cyclic/feedback membership graphs in resonance/looping systems; not assumed as a global ontic axiom in the mainline. Two options are given in Appendix B: (i) final coalgebra semantics for a stratified powerset endofunctor (canonical non-well-founded models), and (ii) NFU-style model notes.

Falsifiability & tests (sketch). To potentially falsify strong (O): find a class of ontic sets provably non-approximable by any convergent family of stratified-parameter protocols relative to $\{\mathcal{A}_i\}$. Candidates: (i) show required parameters are non-measurable in every $\mathcal{A}_i$; or (ii) any protocol requires a non-finite schema, violating the in-principle condition.

B Appendix II — Alternative comprehension schemes (expanded)

Purpose. This appendix expands the brief sketch in the main text comparing two formal defenses against naive comprehension: (A) *measurement-level/operational comprehension* (the REH default) and (B) *syntactic stratification* (Quine-style NF/NFU). We summarize technical properties, model constructions, known consistency results, advantages, disadvantages, and practical recommendations for REH.

B.1 Background — the comprehension problem

Naive set comprehension $\{\, x \mid \varphi(x) \,\}$ for arbitrary formulas φ yields paradoxes (e.g., Russell, Burali–Forti) when applied unrestrictedly. Any robust foundation must restrict comprehension. Two broad strategies are:

- **Operational/semantic restriction (REH default):** admit sets only when their defining predicates are *operationalizable* by measurement/partitions/observable algebras $\mathcal{A}_o$, or explicitly externalized via an agent interpretation I_i with testable membership.
- **Syntactic typing/stratification (NF family):** restrict admissible φ to those that admit a type/stratification assignment.

Both avoid naive comprehension but via different mechanisms: one is epistemic/operational, the other syntactic/ontological.

B.2 Stratified comprehension: NF and NFU (overview)

B.2a What is stratification?

A formula φ is *stratified* if one can assign an integer type $\tau(\cdot)$ to every variable so that

- for each atomic $u \in v$ in φ, $\tau(u) + 1 = \tau(v)$; and
- for each atomic $u = v$, $\tau(u) = \tau(v)$.

Membership raises type by one while equality preserves type. Self-membership $x \in x$ is disallowed since it would require $\tau(x) + 1 = \tau(x)$.

B.2b New Foundations (NF).

Quine's NF adopts *stratified comprehension*: for every stratified $\varphi(x)$ there exists a set $\{\, x \mid \varphi(x) \,\}$. NF admits a *universal set* (e.g., $\{x \mid x = x\}$) and many classical constructions when phrased stratifiably. The (absolute) consistency of NF relative to ZF remains an open problem in set theory.

B.2c NFU (NF with urelements).

NFU is a variant of NF allowing *urelements* (atoms that are not sets). Jensen established a relative consistency result placing NFU on firmer footing via standard set-theoretic methods with mild additional assumptions. Urelements weaken certain combinatorial obstacles and enable model constructions.

B.2d Key technical features.

- **Universal set exists:** $\{x \mid x = x\}$ is a set; class/set distinctions shift relative to ZF.
- **Stratified formulations:** many familiar operations must be phrased stratifiably; some diagonal arguments at the universal level are blocked as the defining formulas are unstratified.
- **Cardinal behavior:** power-set and cardinal arithmetic behave differently than in ZF; comparisons can be stratified/relative.

B.2e Models and constructions.

NFU models use permutation/quotient techniques with urelements; these underlie Jensen's relative consistency proof. NF model building is subtler due to the open consistency question.

B.3 Measurement-level (operational) comprehension — the REH default

B.3a What it says.

Only admit comprehension instances $\{\, x \in U \mid \varphi(x) \,\}$ when φ is *operationalizable*: there exists an observer o and observable algebra $\mathcal{A}_o$ (or measurement map m_o) such that φ corresponds to a measurable predicate in $\mathcal{A}_o$, or φ is explicitly definable via an agent's interpretation I_i together with externalizable membership tests.

B.3b Practical rationale.

This ties set-existence to empirical testability. It is conservative: most applied-math sets used in physics are operational (measurable/definable/computable). Pathological non-empirical comprehensions are blocked without heavy syntactic machinery.

B.3c Technical consequences.

- **Paradox quarantine:** Russell-like self-reference remains purely syntactic unless the predicate is made measurable; there is no ontic image in any $\mathcal{A}_o$ without externalization.
- **Natural blend with applications:** sets correspond to physically accessible classes; continuum tools are allowed as modelling but must respect measurability/tolerance.

B.3d Limits and critiques.

Epistemically conservative and pluralistic: different observers may admit different comprehensions, complicating a single global ontology. For purely mathematical projects seeking a single absolute universe (every definable collection is a set), this approach is restrictive.

B.4 Comparison: NF/NFU vs measurement-level comprehension

- **Ontological vs epistemic restriction.** NF/NFU restricts comprehension ontologically by a syntactic criterion and asserts a global universe (often with a universal set). Measurement-level comprehension restricts *epistemically*—sets exist insofar as they are operationally definable.
- **Paradox control.** Stratification makes the Russell formula unformable; operational comprehension prevents mapping the Russell predicate into observables unless externalized.
- **Consistency/metatheory.** NFU has a strong relative-consistency result (Jensen). NF's consistency is unresolved. Measurement-level comprehension is a methodology layered over a chosen base logic (ZF/NBG/type theory/category-theoretic); its meta-consistency follows the base.
- **Convenience vs fidelity.** NF/NFU offers a single-universe formalism (useful for some theoretical constructions); operational comprehension better matches empirical science and avoids semantic slippage on externalization.

B.5 Recommendations for REH (practical & theoretical)

1. **Default to operational comprehension** in the main body. This preserves the Root Principle and aligns with empirical practice. Use the appendix to record alternatives.
2. If an argument *needs* a universal set or heavy class manipulation, consider:
   (a) work internally with stratified formulations (NF-style) as modelling tools; require an *operational warrant* before externalization, or
   (b) use a conservative class theory (e.g., NBG/MK) and explicitly mark class-level steps, reducing any ontic claim to operational comprehension before it leaves the text.
3. **Use NFU as a mathematical playground** when a consistent system with a universal set is convenient; flag the ontology differences (urelements, altered cardinal arithmetic) and translation costs.
4. For physics modelling, **prefer measurement-level definitions** and tolerance-aware set formation. Use stratified/NF-style comprehension only for internal reasoning kept explicitly un-externalized until an operational test is given.
5. **Document every comprehension instance**: annotate $\{x \mid \varphi(x)\}$ as [OP] (operational/measurable), [STRAT] (stratified), or [NF/NFU-internal]. This keeps proofs auditable and prevents smuggling.

B.6 Further reading (select)

- W. V. Quine, *New Foundations* (original stratified comprehension proposal).
- R. B. Jensen, relative consistency results for NFU.
- P. Aczel, *Non-Well-Founded Sets* (AFA; circular/self-referential membership).
- Standard references on ZF and class theories (NBG/MK) for contrasting comprehension regimes.